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RESEARCH

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War-related traumatic brain injury in Gaza: a multi-center prospective analysis of patterns and outcomes

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Abstract

Background Since October 2023, the Gaza War has caused thousands to suffer from war-related traumatic brain injuries (TBIs) amid the collapse of Gaza's healthcare system. Little is known about the epidemiology and outcomes of TBI in such severely resource-limited settings.

Methods We conducted a prospective cohort study at the two largest neurosurgical centers in the southern Gaza Strip from July 15, 2024, to January 19, 2025. Patients with war-related TBI were enrolled consecutively and followed for 30 days after admission. Data were collected on demographics, clinical presentation, imaging findings, interventions, complications, and outcomes. The primary outcome was 30-day mortality, while secondary outcomes included the neurological status at discharge and complications.

Results A total of 244 patients were included, with a median age of 21 years, and 74.5% were males. The 30-day mortality rate was 26.2%. Severe TBI (GCS ≤ 8) at admission was associated with a higher mortality compared to mild and moderate TBI ($p < 0.001$). Non-survivors had significantly higher rates of multilobar and bilateral injuries, subdural and intraventricular hemorrhages, midline shifts, and effaced basal cisterns. Among survivors (n = 180), 27.2% experienced neurological deficits at discharge, most commonly motor impairment and aphasia. Neurological deficits were linked to penetrating injuries, multilobar involvement, midline shifts, and ≥ 3 shrapnel fragments on imaging. Complication rates were generally low but higher among individuals with neurological impairments. Multivariate regression analysis showed that TBI severity (moderate: aRR = 7.05, 95% CI: 2.32–14.23; severe: aRR = 9.91, 95% CI: 4.56–18.64), older age (aRR = 1.02 per year, 95% CI: 1.01–1.03), brain matter extrusion (aRR = 2.24, 95% CI: 1.06–4.70), intraventricular hemorrhage (aRR = 2.38, 95% CI: 1.39–4.03), and subdural hemorrhage (aRR = 1.90, 95% CI: 1.30–4.03) were significant predictors of 30-day mortality.

Conclusion In this cohort of war-related TBI patients in Gaza, mortality was significantly linked to admission GCS, age, brain matter extrusion, IVH, and SDH. The study highlights how resource-limited, conflict-driven healthcare disruptions impact TBI outcomes and emphasizes the need to strengthen neurosurgical capacity, emergency response systems, and rehabilitation efforts in such environments.

Keywords Armed conflicts, Traumatic brain injuries, Gaza strip, Health services accessibility, Prospective studies, Survival rate

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Background

The Gaza Strip is a densely populated, resource-limited area under Israeli occupation since 1967. Decades of conflict, political instability, a two-decade-long blockade, and repeated military assaults have created conditions of chronic vulnerability and scarcity, leaving healthcare services severely overstretched and under-resourced [1]. The war in Gaza, ongoing since October 7, 2023, is one of the most intense and destructive conflicts in the region's history. By September 2025, over 63,000 Palestinians had been killed, and more than twice as many wounded, with many suffering from permanent disabilities such as traumatic brain injuries, spinal cord injuries, and limb amputations [2, 3]. In addition to the direct casualties, the war has inflicted severe humanitarian suffering, including extreme food insecurity, the forced displacement of over 90% of the population, destruction of infrastructure, inaccessibility to aid and medical care, repeated attacks on health facilities and personnel, movement restrictions, inadequate sanitation, and widespread psychological trauma [4–7].

The destruction of healthcare infrastructure has been particularly devastating. Many facilities are now non-functional, and severe shortages of medications, surgical equipment, clean water, and generator fuel have crippled those still operating [4, 8]. In May 2025, only 22 out of 36 hospitals were partially functional, and fewer than half of Gaza's primary care clinics remained open [4]. Meanwhile, more than a thousand healthcare workers have been killed, and the remaining staff are forced to work in dangerous and exhausting conditions [9, 10].

Blast-related traumatic brain injuries (TBIs) can be classified as primary (caused by blast waves), secondary (caused by projectiles such as shrapnel and bullets), tertiary (resulting from head acceleration and impact), and quaternary (stemming from other effects like burns, inhalation injuries, or crush trauma) [11]. These mechanisms can lead to skull fractures, intracranial hemorrhages, and diffuse axonal injury [12], all of which are associated with poor prognoses and are difficult to manage, even in advanced medical systems.

Historically, TBIs have been a major cause of morbidity and mortality in war zones [13]. During World War I, head injuries accounted for up to 25% of battlefield trauma, leading to the development of cranioplasty and wound care techniques [14]. In more recent conflicts like Iraq and Afghanistan, the use of advanced explosives has created more complex TBI patterns and driven innovations such as damage control neurosurgery, which focuses on early decompression and hemorrhage management [13, 15]. However, such procedures are rarely practical in low-resource civilian war settings, like the Gaza conflict, where 72% of TBI-related deaths reportedly occur before reaching a hospital [15, 16].

It is estimated that 20% of the wounded require long-term rehabilitation [2]. The ongoing blockade has further hindered referrals for specialized care, forcing hospitals to operate under siege-like conditions: overwhelmed, understaffed, and lacking essential resources. Despite the widespread trauma among Gaza's civilian population, data on the epidemiology, management, and outcomes of TBIs in such high-intensity, low-resource conflict settings remain very limited. This study aims to address this knowledge gap by systematically analyzing the patterns and outcomes of TBIs sustained over a six-month period during the Gaza War.

Methods

Design, participants, and operational definitions

This prospective observational cohort study was conducted from July 15, 2024, to January 19, 2025. Eligible participants included patients of any age and sex who were admitted to the neurosurgical departments at the European Gaza Hospital (EGH) and Nasser Medical Complex (NMC) due to war-related TBI.

Patients with a Glasgow Coma Scale (GCS) score of 3 and bilaterally fixed, dilated pupils were classified as "code black" at triage. They typically died shortly after arriving at the emergency department. Due to the collapse of medical documentation systems under wartime conditions and the overwhelming patient load, these cases were not reliably recorded, leading to significant missing data. Consequently, they could not be included in the study cohort. The study also excluded individuals with injuries unrelated to the war.

War-related Traumatic Brain Injury (TBI) was defined as brain damage secondary to an externally inflicted trauma such as explosive blasts, blunt trauma, or penetrating injuries [17].

TBI severity was classified based on the presenting GCS as follows: mild (GCS scores of 13–15), moderate (GCS scores of 9–12), or severe (GCS scores ≤ 8) [18, 19].

Concomitant injuries were defined as those that required hospital admission on their own merit, independent of the TBI.

Complications included adverse events that occurred during hospitalization and up to 30 days after admission, such as brain abscess, wound infections, meningitis, CSF leaks, hydrocephalus, and rebleeding.

Glasgow Outcome Scale (GOS) was categorized on a scale from one to five as follows: one = death, two = persistent vegetative state, three = severe disability (dependence on others for daily support), four = moderate disability (independent in daily life but with some

residual deficits that interfere with complex activities), five = good recovery (resumed normal activities even if some minor neurological or psychological deficits may still be present) [20].

Neurological deficits upon discharge refer to new focal or generalized neurological impairments identified at the time of hospital discharge, and they are classified into the following categories: language deficits, unsteadiness, visual deficits, impaired consciousness, facial asymmetry, irritability/confusion, sensory deficits, and motor deficits.

Study settings and trauma care infrastructure

Before the current war, Gaza's neurosurgical services were already strained by ongoing shortages, weak infrastructure, and limited ICU capacity. Two main departments at Shifa Medical Complex and EGH provided essential neurosurgical care, staffed by four board-certified neurosurgeons, seven master's-level physicians, and 15 residents, and hosted the local neurosurgery board program [21, 22]. Despite these challenges, these centers performed procedures such as VP shunts, tumor resections, spinal surgeries, and emergency craniotomies with outcomes approaching international standards [21, 22].

The war profoundly disrupted these services. Shifa Medical Complex was destroyed early in 2024, and EGH was attacked and later occupied, becoming inoperable since May 2025 [23]. By mid-2025, only two board-certified neurosurgeons, four master's-level physicians, and 12 residents remained in Gaza. Neurosurgical care became fragmented and mainly focused on treating war injuries. Israeli military operations also split Gaza into two isolated zones, north and south, making medical referrals between them impossible [8]. The neurosurgical departments featured in this study represented the most advanced units in the southern enclave at that time, located at EGH and Nasser Medical Complex (NMC). EGH hosted the last functional board training program and employed the only two remaining board-certified neurosurgeons in the Strip, along with eight residents and two master's-level specialists.

Prehospital care for trauma patients was nearly nonexistent due to the collapse of ambulance services and ongoing hostilities; most patients were transported by civilians without stabilization. In the emergency department, rapid triage was performed under mass casualty conditions, with immediate focus on airway management, bleeding control, and neurological assessment. Intubation was attempted when feasible, but ventilators were scarce and limited to ICU settings, leaving many patients dependent on manual bag-valve ventilation while waiting for ICU beds. During the study period, only 40 ventilator-equipped ICU beds were operational across the Strip, serving a population of over two million people [8]. Invasive ICP monitoring was unavailable, so ICP elevation was assessed clinically and through imaging. Tranexamic acid was not available, and osmotic agents (mannitol or hypertonic saline) were inconsistently accessible due to supply chain disruptions.

Upon admission, all TBI patients received standardized medical management: empirical antibiotic prophylaxis with ceftriaxone 1 g IV twice daily for adults (or 40 mg/kg for children), antiseizure prophylaxis with levetiracetam 500 mg twice daily or phenytoin 5 mg/kg, and omeprazole 40 mg daily (or 1 mg/kg/day for pediatric patients) for GI protection, along with analgesia. In cases of penetrating TBI, vancomycin 15 mg/kg IV was also administered.

Surgical decisions were made based on radiological findings and the patient's clinical status. Decompressive craniectomy was performed for elevated ICP, acute subdural hematoma, and > 5 mm midline shift, with the dura left open to reduce pressure. In cases of epidural or extra-axial hematomas without significant edema, craniotomy and hematoma evacuation were carried out. Debridement with bone elevation, with or without dural repair, was used for depressed skull fractures with brain herniation, and EVD insertion was performed in cases of hydrocephalus.

Patients were discharged either to inpatient rehabilitation, which is available only at a single center in Deir Albalah in the Middle Governorate with very limited capacity, or to outpatient physiotherapy, depending on service availability and injury severity. Due to the severe deterioration of rehabilitation infrastructure in Gaza [2], many were sent home without structured follow-up, raising the risk of long-term disability and impaired functional recovery.

Outcomes

The primary outcome of the study was the mortality rate within 30 days of admission. Secondary outcomes included neurological status at discharge, classified by the GOS and GCS scores, the presence of neurological deficits, and the rates of surgical complications.

Data collection

A trained fourth-year neurosurgery resident (the first author) collected the data using a standardized, Excel-based case reporting form. Data was gathered from bedside observations as well as from medical files, operative notes, and radiology reports. The information obtained at admission included patient demographics, such as age and sex, clinical data on presentation, including the chief presenting symptom, mechanism(s) of injury, admission GCS scores, computed tomography (CT) findings from

formal radiology reports (MRI was unavailable), and the presence of concurrent injuries, along with laboratory parameters like hemoglobin level upon admission.

Collected information related to the clinical course included the need for ICU admission, total ICU and hospital stay durations, the requirement for surgical intervention, the types of interventions performed, and the occurrence of complications.

The 30-day outcome data (survival) were gathered during hospitalization and from neurosurgical outpatient clinic visits, which took place in a high-volume outpatient neurosurgical clinic with an average caseload of approximately 120 patients per session. Patients who missed their clinic appointments were contacted by phone to determine the primary outcome.

The secondary outcome (neurological deficits) was recorded at discharge and included the GOS scores, GCS scores, and the presence of neurological deficits. The other secondary outcome (complications) was documented during admission and at the 30-day outpatient visit. For survivors who did not attend the outpatient clinic, complications were reported up to the day of discharge. Figure 1 shows key milestones for data collection and outcome reporting.

Ethical considerations

The study was conducted following the principles outlined in the Declaration of Helsinki. Ethical approval was granted by the Research Department at the Ministry of Health in Gaza. Additionally, the Institutional Review Board (IRB) at the Islamic University of Gaza approved the study (approval letter number: 2024/06). Informed consent was obtained directly from patients or, for those with impaired consciousness, from their next of kin. All patient information was anonymized and securely stored. The methods and results reporting adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Checklist [24].

Statistical analysis

Descriptive statistics summarized the data: the shape and distribution were analyzed to report either the mean and standard deviation or the median and interquartile range for continuous variables, as appropriate; categorical variables were summarized as frequency and percentage.

Univariate comparisons were conducted using the Mann–Whitney U test, Student's t-test, chi-squared test, and Fisher's exact test, as appropriate. Multivariable modified Poisson regression with robust error variances was employed to estimate adjusted relative risks (RRs) for the primary outcome. The model was adjusted for all relevant clinical covariates, including demographics, clinical features, and radiological findings. Basal cistern status, midline shift, and shrapnel presence were excluded from the model due to a high percentage of missing data (>40%). A p-value below 0.05 was considered statistically significant. All analyses were performed using R software.

Results

Baseline sociodemographic and clinical characteristics

A total of 244 patients with war-related traumatic brain injury (TBI) were admitted over the 6-month study period, and all were included. The median age of the cohort was 21 years (IQR: 12–32), and 74.5% were male (Table 1).

The most common presenting symptom was decreased level of consciousness (LOC), observed in 64.3% of cases, and it was significantly more frequent in non-survivors (98.4% vs. 52.2%, $p<0.001$). Conversely, headache (32.2% vs. 1.6%, $p<0.001$) and repeated vomiting (10.6% vs. 0%, $p=0.004$) were more common among survivors. Overall, 105 patients with severe TBI (95.5%), nine patients with moderate TBI (37.5%), and 14 patients with mild TBI (12.7%) were admitted to the ICU. Of the five patients with severe TBI who were not admitted to the ICU, three died while two survived to the one-month follow-up. Non-survivors were more likely to have been admitted to the ICU (90.6% vs. 38.9%, $p<0.001$), although the duration of ICU stay did not differ significantly between the two groups. Conservative management was the primary approach (71.7%). Severe TBI (GCS 3–8) was significantly more frequent in non-survivors (87.5% vs. 30%, $p<0.001$) (Table 1).

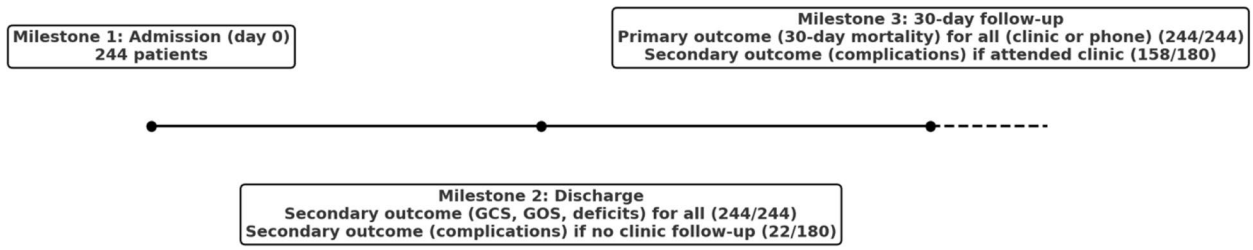

Fig. 1 Timeline of Outcome Assessment for Patients with TBI in the Cohort

Table 1 Demographic and clinical characteristics of TBI patients, stratified by survival outcome ($N = 244$)

Variable	N	Total cohort ($n = 244$)	Survivors ($n = 180$)	Non-survivors ($n = 64$)	*p*-value
Age, years, median (IQR)	244	21 (12–32)	20 (10–31)	26.5 (15.75–36.25)	0.01
Male, n (%)	244	182 (74.5)	131 (72.8)	51 (79.7)	0.27
Mechanism of injury, n (%)	244				0.14
Penetrating injury		132 (54.1)	92 (51.1)	40 (62.5)	
Blunt force		112 (45.9)	88 (48.9)	24 (37.5)	
Gunshot wound, n (%)	235[a]	5 (2.0)	2 (1.1)	3 (4.9)	0.11
GCS category, n (%)	244				< 0.001
Mild (13–15)		110 (45.1)	108 (60.0)	2 (3.1)	
Moderate (9–12)		24 (9.8)	18 (10.0)	6 (9.4)	
Severe (3–8)		110 (45.1)	54 (30.0)	56 (87.5)	
Anisocoria, n (%)	235[a]	6 (2.5)	3 (1.7)	3 (4.9)	0.18
Chief presenting symptom, n (%)	244				
Seizure(s)		1 (0.4)	1 (0.6)	0 (0.0)	1.00
Decreased LOC		157 (64.3)	94 (52.2)	63 (98.4)	< 0.001
Headache		59 (24.2)	58 (32.2)	1 (1.6)	< 0.001
Repeated vomiting		19 (7.8)	19 (10.6)	0 (0.0)	0.004
Weakness		8 (3.3)	8 (4.4)	0 (0.0)	0.11
Concomitant injuries, n (%)	244	48 (19.7)	12 (18.8)	36 (20)	0.82
Haemoglobin, g/dL, mean (SD)	224[a]	11.3 (1.9)	11.3 (1.9)	11.4 (2.0)	0.74
Management style, n (%)	244				0.27
Surgical		69 (28.3)	47 (26.1)	22 (34.4)	
Conservative		175 (71.7)	133 (73.9)	42 (65.6)	
Surgical Intervention, n (%)	69				
Craniectomy		24 (9.8)	12 (6.7)	12 (18.8)	
Wound debridement		19 (7.8)	12 (6.7)	7 (10.9)	
Craniotomy with bone flap elevation		15 (6.1)	15 (8.3)	0 (0)	
EVD/shunt placement		2 (0.8)	0 (0)	2 (3.1)	
Hematoma evacuation		7 (2.9)	6 (3.3)	1 (1.6)	
Bullet removal		2 (0.8)	2 (1.1)	0 (0)	
ICU admission, n (%)	244	128 (52.5)	70 (38.9)	58 (90.6)	< 0.001
ICU stay, days, median (IQR)	128	4 (2–8)	4 (2–8)	2.5 (1.75–3.25)	0.23
Hospital stay (days), median (IQR)	232[a]	5 (2–9)	5 (2–8)	5 (3–10)	0.26

[a] Represent variables with missing data for a few cases. Percentages are calculated from available data

Surgical interventions were carried out in 28.3% of patients. Craniotomies with bone flap elevation were performed exclusively in survivors (6.1%, $p = 0.01$). Other procedures showed no significant differences between groups.

Radiological Findings

Non-survivors exhibited significantly higher rates of multi-lobar (65.6% vs. 35%, $p = 0.001$) and bilateral (37.5% vs. 16.7%, $p = 0.002$) injuries compared to survivors. Subdural hemorrhage, intraventricular hemorrhage, effaced basal cisterns, and midline shift were all significantly more common among non-survivors ($p < 0.05$ for each). Conversely, epidural hematomas ($p = 0.008$) and depressed skull fractures ($p = 0.01$) were observed more frequently in survivors. Other findings, such as cerebral contusion, intracerebral hemorrhage, and shrapnel presence, did not show significant differences (Supplementary Table 1).

Among survivors, radiological findings linked to neurological deficits included multi-lobar injuries (51% vs. 29%, $p = 0.001$), bilateral involvement (36.7% vs. 9.2%, $p < 0.001$), intraventricular hemorrhage (10.2% vs. 2.3%, $p = 0.03$), and midline shift (51.9% vs. 25.8%, $p = 0.02$). Conversely, frontal lobe injuries and depressed skull fractures were more common among survivors without deficits ($p < 0.05$). Patients with ≥ 3 shrapnel

fragments were more likely to have neurological deficits ($p = 0.005$) (Supplementary Table 2).

Primary Outcome and Secondary Outcomes

The 30-day mortality rate was 26.2%., with all mortalities occurring during hospital admission (Table 1). Non-survivors had a higher incidence of severe injury patterns both clinically and radiologically, as detailed in Table 1 and Supplementary Table 1.

Complications occurred in 11.1% of patients, including wound infections (5.3%) and brain abscesses (2.9%), with no significant differences in complication rates between survivors and non-survivors (Table 2).

Among survivors, 27.2% were discharged with neurological deficits (Tables 3 and 4). Those with deficits had longer hospital stays (median (IQR) 10 (7, 17.75) vs. 4 [2, 7] days, $p < 0.001$), higher ICU admission rates (67.3% vs. 28.2%, $p < 0.001$), and were more likely to have severe TBI (53.1% vs. 21.4%, $p < 0.001$). Decreased LOC and weakness at presentation were linked to deficits ($p < 0.001$ for both), while headache and repeated vomiting were more common in those without deficits ($p < 0.05$). Penetrating injuries occurred significantly more often in patients with neurological deficits (71.4% vs. 43.5%, $p < 0.001$). Rates of surgical management and other complications were similar between groups (Table 3).

Table 4 shows survivor discharge outcomes, including GCS scores and neurological deficits. Most survivors (90.6%) had GCS scores of 13–15, while only two (1.1%) had a GCS below eight and were discharged to rehabilitation in an unresponsive wakefulness state. Concerning specific neurological deficits, motor deficits were the most common (20.6%), followed by aphasia (3.9%), and impaired consciousness (2.2%). Figure 2 illustrates the distribution of Glasgow Outcome Scores (GOS) across age groups and injury severities, showing a clear trend toward poorer outcomes with increasing TBI severity, while mild TBI cases predominantly achieved favorable recovery.

Predictors of 30-Day Survival Following War-Related TBI

The Poisson regression analysis identified several significant independent predictors of mortality. Patients with moderate (aRR = 7.05, 95% CI: 2.32–14.23, $P = 0.004$) and severe (aRR = 9.91, 95% CI: 4.56–18.64, $P = 0.002$) GCS scores at admission had notably higher mortality risks compared to those with mild GCS scores. Additionally, older age was linked to increased mortality risk (aRR = 1.02 per year, 95% CI: 1.01–1.03, $P = 0.006$). Brain matter extrusion (aRR = 2.24, 95% CI: 1.06–4.70, $P = 0.03$), intraventricular hemorrhage (aRR = 2.38, 95% CI: 1.39–4.03, $P = 0.001$), and subdural hemorrhage (aRR = 1.90, 95% CI: 1.30–4.03, $P = 0.001$) were also strong predictors of mortality. The other clinical and radiological factors showed no significant association with mortality after adjustment.

Discussion

This study provides a detailed analysis of war-related TBI patterns and outcomes during the Gaza War (2023–), emphasizing the devastating impact of war on populations in low-resource settings. Results show high mortality and morbidity, with only 73.8% of patients surviving the first 30 days after injury, and many experiencing neurological deficits. This is the first study from the Gaza Strip and one of the few similar reports from conflict zones. Additionally, the consecutive sampling of all TBI admissions at two major hospitals over six months improves the data's representativeness and supports credible conclusions.

Males and younger individuals were more likely to be admitted with TBI, aligning with previous conflict-zone studies [13, 25]. A Syrian study found that 79% of 195 TBI victims were male, with a median age of 25 [25], while a Gaza ICU study during the current war reported a median age of 26 among 101 consecutive admissions [8]. The average age in our study was also younger than that of military cohorts (mean, 27 years) [26]. This reflects Gaza's young population and cultural patriarchal norms where males are more exposed due to provider roles [25]. The lack of protective gear and the absence of safe

Table 2 Complications in TBI patients (stratified by survival outcome)

Variable	N[a]	Total cohort	Non-survivors ($n = 64$)	Survivors ($n = 180$)	*p*-value
Brain abscess	27	7 (2.9)	0 (0.0%)	7 (3.9%)	0.19
Wound infection		13 (5.3)	3 (4.7%)	10 (5.6%)	1.00
CSF leak		1 (0.4)	0 (0.0%)	1 (0.6%)	1.00
Hydrocephalus		4 (1.6)	2 (3.1%)	2 (1.1%)	0.28
Meningitis		2 (0.8)	0 (0.0%)	2 (1.1%)	1.00
Rebleeding		3 (1.2)	1 (1.6%)	2 (1.1%)	1.00

[a] Complications are reported up to discharge for 22 survivors who did not attend the 30-day follow up

Table 3 Demographic and clinical characteristics of TBI survivors, stratified by neurological outcome ($N = 180$)

Variable	N	No deficit ($n = 131$)	Neurological deficit ($n = 49$)	*p*-value
Age, years, median (IQR)	180	20 (9.5–29.5)	21 (11–36)	0.23
Male, n (%)	180	98 (74.8)	33 (67.3)	0.31
Mechanism of injury, n (%)	180			< 0.001
Penetrating injury		57 (43.5)	35 (71.4)	
Blunt force		74 (56.5)	14 (28.6)	
Gunshot wound, n (%)	174*	1 (0.8)	1 (2.1)	0.47
GCS category, n (%)	180			< 0.001
Mild (13–15)		91 (69.5)	17 (34.7)	
Moderate (9–12)		12 (9.2)	6 (12.2)	
Severe (3–8)		28 (21.4)	26 (53.1)	
Anisocoria, n (%)	174*	2 (1.6)	1 (2.1)	1.00
Chief presenting symptom, n (%)	180			
Seizure(s)		1 (0.8)	0 (0)	1.00
Decreased LOC		56 (42.7)	38 (77.6)	< 0.001
Headache		56 (42.7)	2 (4.1)	< 0.001
Repeated vomiting		18 (13.7)	1 (2)	0.02
Weakness		0 (0)	8 (16.3)	< 0.001
Concomitant injuries, n (%)	180	30 (22.9)	6 (12.2)	0.11
Haemoglobin, g/dL, mean (SD)	167*	11.2 (1.85)	11.4 (2)	0.64
Management style, n (%)	180			0.34
Surgical		31 (23.7)	15 (30.6)	
Conservative		100 (76.3)	34 (69.4)	
ICU admission, n (%)	180	37 (28.2)	33 (67.3)	< 0.001
ICU stay, days, median (IQR)	63*	3 (2, 6)	5 (2, 8)	0.29
Hospital stay (days), median (IQR)	169*	4 (2,7)	10 (7, 17.75)	< 0.001
Complication(s)	180	13 (9.9)	10 (20.4)	0.06
Brain abscess		4 (3.1%)	3 (6.1%)	0.39
Wound infection		7 (5.3%)	3 (6.1%)	1.00
CSF leak		0 (0%)	1 (2%)	0.27
Hydrocephalus		1 (0.8%)	1 (2%)	0.47
Meningitis		1 (0.8%)	1 (2%)	0.47
Rebleeding		1 (0.8%)	1 (2%)	0.47

*Indicates missing data for some patients

zones and corridors further increase these risks [27]. Another possible factor is that younger patients are often given priority during triage for limited ICU and surgical resources. Although hospitals in Gaza do not practice formal withdrawal of care, this age-based prioritization, influenced by resource shortages, may have affected the patterns of admission and survival observed.

On the other hand, while males were overrepresented among those with TBI, women and children together represented nearly half the cohort, reflecting casualty reports where women, children, and the elderly collectively account for an estimated 60% of traumatic deaths during the Gaza war [16]. This highlights the nature of the warfare in Gaza, where aerial bombardment and artillery shelling, rather than small-arms fire, were the main modes of attack, often killing entire families within the targeted structures and injuring those nearby—who are more likely to be males [28].

The severity of TBI, as indicated by the admission GCS, was a significant predictor of mortality. Severe TBI (GCS ≤ 8) was present in 87.5% of non-survivors in our cohort, while moderate and severe TBI had seven and ten times higher probabilities of death, respectively, compared to those with mild TBI. This aligns with the literature highlighting the strong association between TBI severity and mortality [13, 15, 29, 30]. Similarly, in the ICU cohort from Gaza mentioned earlier, intracranial mass effect was a significant factor associated with ICU mortality [8].

Table 4 Discharge characteristics of the cohort and neurological deficits among survivors

Outcome Measure	Frequency (%)	Population
GCS Category at Discharge		Survivors only (*N* = 180)
Mild (13–15)	163 (90.6%)	
Moderate (9–12)	13 (7.2%)	
Severe (3–8)	4 (2.2%)	
GOS Category at Discharge		Full cohort (*N* = 244)
One (death)	64 (26.2%)	
Two (persistent vegetative state)	5 (2%)	
Three (severe disability)	20 (8.2%)	
Four (moderate disability)	21 (8.6%)	
Five (good recovery)	134 (54.9%)	
Neurological Deficits		Survivors only (*N* = 180)
Language deficits (aphasia)	7 (3.9%)	
Unsteadiness	1 (0.6%)	
Visual deficits	1 (0.6%)	
Impaired consciousness (unresponsive wakefulness or a minimally conscious state)	4 (2.2%)	
Facial asymmetry	1 (0.6%)	
Irritability/confusion	2 (1.1%)	
Sensory deficits	1 (0.6%)	
Motor deficits	37 (20.6%)	

one = death, two = persistent vegetative state, three = severe disability (dependence on others for daily support), four = moderate disability (independent in daily life but with some residual deficits that interfere with complex activities), five = good recovery

Radiological findings revealed a strong association between injury severity and mortality. Non-survivors exhibited higher rates of multilobar and bihemispheric injuries, subdural hemorrhage (SDH), intraventricular hemorrhage (IVH), midline shift, and effaced basal cisterns—findings consistent with previous TBI studies in Iraq, Afghanistan, and Latin America [11, 12, 29]. Survivors, however, more frequently presented with epidural hematomas and depressed skull fractures, which may be more amenable to surgical intervention even in low-resource settings. SDHs often occur alongside parenchymal damage (e.g., contusions, edema, diffuse axonal injury), contributing to a poorer prognosis [31].

Compared to other war environments, the current group showed significantly higher rates of multilobar injuries (43%), SDH (22.5%), and IVH (10.7%) than groups from Syria or U.S. military data [11, 25]. Midline shift was present in 42% of patients with available radiological data and was notably more common in non-survivors ($p = 0.003$). Bihemispheric injuries (22%) were also more frequent among non-survivors. In comparison, a midline shift was reported in 17.7% of the Ukrainian and 18.5% of the Syrian TBI groups, where it also predicted worse outcomes [25, 32]. These findings emphasize the extreme severity of injuries in Gaza, exceeding those documented in other high-intensity conflict zones.

In terms of mortality, 26.2% of our cohort died within 30 days of their injuries, compared to 33% in Syria and 26.9% in Afghanistan [25, 33]. Conversely, the Ukrainian study reported a 30-day mortality rate of 6.4%, while the overall mortality rate for modern combat-related TBIs was 18% in a 2022 meta-analysis [26, 32]. Methodological differences might explain some of the discrepancy, but these observations also highlight the severity of violence in the Gaza war and the critical state of healthcare services in low-income, chronic conflict regions like the Gaza Strip. However, the reported mortality rate likely reflects selection bias by excluding the most critical cases (e.g., GCS 3 with fixed pupils) who died before documentation. Prehospital deaths were also not included. The young median age (21 years), triage practices that may favor better prognoses, and centralized neurosurgical care in two operational hospitals may also have contributed to the relatively favorable survival rate among those who reached definitive care.

Among survivors, 27.2% were discharged with neurological impairments, mainly motor deficits and aphasia, similar to the 25.1% reported in the Syrian cohort [25]. These findings emphasize the significant burden of long-term disability after war-related TBI. The destruction of Gaza's rehabilitation infrastructure further endangers survivors' recovery and reintegration [2]. Neurological impairment was strongly associated with penetrating

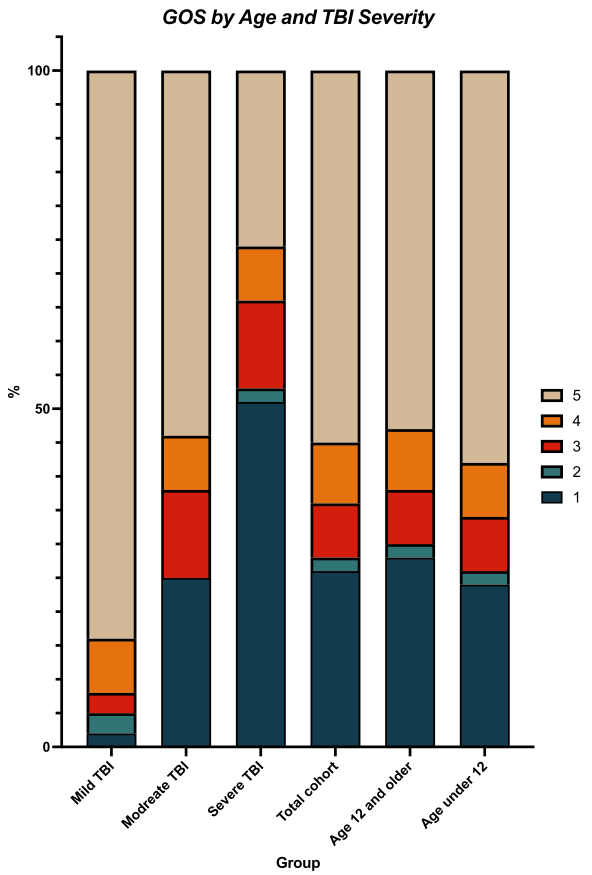

Fig. 2 Distribution of discharge Glasgow Outcome Scores by Age Group and Presentation Severity

injuries, multilobar and bilateral damage, and midline shift. Likewise, the Syrian cohort found that lower admission GCS was linked to post-discharge deficits [25]. Other long-term effects, such as epilepsy, cognitive decline, and behavioral changes, have been observed in extended follow-ups and warrant further investigation [29, 34, 35].

Complication rates in our cohort were lower than those in other civilian war zones but higher than in military settings. Wound infections (5.3%) and brain abscesses (2.9%) occurred more often than in U.S. military populations, where early evacuation and sterile conditions help reduce infections [36]. Conversely, infectious complications affected 10.5% of our cohort, similar to a cohort of Syrian combatants treated in Israel (10.6%) [37]. However, these rates were much lower than in a Syrian cohort where brain abscesses affected 18% of patients. This difference may be due to the use of dual antibiotic coverage in Gaza for penetrating injuries or differences in local antimicrobial resistance and hygienic practices [25]. The short follow-up period in this report might also have influenced the results. Hydrocephalus was noted in 1.6% of patients, slightly below rates seen in other civilian settings.

The surgical intervention rate (27.9%) in this cohort was significantly lower than the 40–60% reported in trauma centers in high-income countries [38]. Craniectomy was performed in 7.8% of cases, half the rate observed in U.S. military settings (15–20%) [15], and bone flap elevation was only done in survivors. Studies have associated surgical management with improved survival in patients with TBI from cranial gunshot wounds [39]. Although including those with mild TBI may have influenced this finding, these rates are also likely affected by operational challenges during wartime, such as unstable power supply, constantly busy operating rooms, high casualty numbers, and a critical shortage of neurosurgeons and critical care resources [8, 21, 40]. For example, during this period, only two board-certified neurosurgeons served Gaza's two million residents (a ratio of one per million).

The findings of this study highlight the urgent need for coordinated policies to improve neurotrauma care in conflict zones. Investments should focus on equipping hospitals with neurosurgical and intensive care capabilities, ensuring a steady supply of essential medical and surgical resources, and training local staff in advanced trauma and neurocritical care. Moreover, systematic rehabilitation programs for survivors with neurological disabilities should be incorporated into post-acute care to enhance functional outcomes. On the international level, protecting healthcare infrastructure and personnel, as mandated by humanitarian law, remains vital to ensuring the delivery of timely and effective care to victims of war-related TBI.

The study has several limitations. First, data from the isolated northern part of the Gaza Strip were unavailable, which at the time had fewer neurosurgical resources. Also, patients presenting with GCS 3 and bilaterally fixed pupils (usually triaged as "code black") were not included due to rapid demise and lack of documentation. These factors introduced selection bias that likely led to an underestimation of the overall TBI mortality and morbidity burden. Second, long-term follow-up of patients was not feasible, possibly causing an underestimation of late complications like hydrocephalus or delayed neurological deterioration. Third, radiological data were occasionally incomplete and lacked advanced imaging, such as MRI, which limited the ability to identify subtle injuries, including diffuse axonal injury. Fourth, although the study aimed to assess injury severity, head AIS scores could not be provided because of the absence of coding systems and trauma registries during emergency conditions. Lastly, despite prospective data collection, resource limitations and security risks may have affected the ability to control for all confounding variables.

Conclusions

This prospective two-center study highlights the substantial burden of war-related traumatic brain injuries (TBI) in the Gaza Strip, with high rates of mortality and neurological disability. Nearly one in four patients died, and over a quarter of survivors were discharged with neurological deficits. Worse outcomes were linked to older age, severe TBI, decreased level of consciousness at presentation, multi-lobar and bilateral injuries, and specific radiological findings. In comparison, epidural hematomas and frontal lobe injuries were more common among those with better outcomes.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions
BAJ: conceptualization and study design. All authors contributed to the literature review and tool development. BAJ contributed to and supervised data collection. All authors contributed to the data analysis and interpretation, as well as to the writing of the manuscript. All authors approved the final version.

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Data availability
Available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Ethical approval for this study was obtained from the Institutional Review Board (IRB) at the Islamic University of Gaza. Participants or their next of kin provided informed written consent. The confidentiality of the data was maintained throughout the data collection and analysis process.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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